

Theory of Machine Elements

Detailed Lecture-wise Break-up of Course Content

Lecture Number (1 h Each)	Content
1	Introduction to theory of machine elements
2,3	Mechanisms and machines
4,5	Kinematic links, joints, kinematic pairs, kinematic chains, mechanisms and structures;
6,7,8	Degrees of freedom, determination of degrees of freedom, Grublers and Kutzbach criteria, Paradoxes
9,10	Grashof Condition, Inversions of four-bar chain, single-slider and double-slider crank chains
11, 12, 13, 14, 15,16	Kinematic Analysis: graphical methods and analytical methods for displacement, velocity, and acceleration
17, 18	Velocity and acceleration analysis of slotted lever mechanism, Coriolis analysis
19,20,21,22,23,24,25	Lower pair mechanisms: Hooke's joint, Belt, rope and chain drives
26,27,28,29,30,31	Higher pair mechanisms: Cam and follower, gear trains
32,33,34	Inertia forces and turning moment diagram
35,36,37,38	Precessional motion and Gyroscopic couple
39,40,41,42	Balancing of rotating and reciprocating masses
43,44,45	Governors